

A Constant-Shift Counterexample to the Unnormalised FCB Tensor-Network Question

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Source and Target Question

The source paper is Francisco Escudero Gutierrez, *Influences of Fourier Completely Bounded Polynomials and Classical Simulation of Quantum Algorithms*, arXiv:2304.06713 [1].

The paper introduces Fourier completely bounded norms for Boolean-cube polynomials and proposes a weaker variant of the Aaronson–Ambainis influence conjecture. At the end of the paper, Question 4.5 asks:

Given a polynomial p of degree at most d , is there $(u, v, A) \in \mathcal{BB}^d$ such that

$$\langle u, A(i_1) \cdots A(i_d) v \rangle = \frac{\hat{p}(S_{\mathbf{i}})}{\text{poly}(d, \text{MaxInf}[p])}$$

for any $\mathbf{i} \in [n+1]^d$?

Remark 4.4. However, there might be a different way of, given a polynomial p of degree at most d , choosing (u, v, A) with Boolean behavior of degree d such that

$$\langle u, A(i_1) \cdots A(i_d) v \rangle = \frac{\hat{p}(S_{\mathbf{i}})}{\text{poly}(d, \text{MaxInf}[p])},$$

for any $\mathbf{i} \in [n+1]^d$. If that was true, one could copy and paste the proof of Theorem 1.6 and conclude Theorem 1.5. This reduces Theorem 1.5 to a question with flavor of tensor networks (at the end of the day we are seeking for an MPS, but caring about the norm of the matrices instead of the dimension) [CPGSV21] and almost-quantum correlations (we could also say that we look for (u, v, A) that define a correlation with some symmetry defined by the Boolean behavior, without imposing the symmetries on the A) [NGHA15].

Question 4.5. *Given a polynomial p of degree at most d , is there $(u, v, A) \in \mathcal{BB}^d$ such that*

$$\langle u, A(i_1) \cdots A(i_d) v \rangle = \frac{\hat{p}(S_{\mathbf{i}})}{\text{poly}(d, \text{MaxInf}[p])},$$

for any $\mathbf{i} \in [n+1]^d$?

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Idea of the Proof

The obstruction is the constant Fourier coefficient. The maximum influence does not see constants: adding M to a polynomial leaves every influence unchanged, but changes $\hat{p}(\emptyset)$ by M . On the

other hand, every degree- d correlation produced by unit vectors and contractions has absolute value at most 1. Therefore no denominator depending only on d and $\text{MaxInf}[p]$ can represent all unnormalised polynomials.

This is a narrow reading of the printed auxiliary question. The normalized conjecture earlier in the source, where $\|p\|_{\text{fcb},d} \leq 1$, is not affected by this counterexample.

Definitions

For a Boolean-cube polynomial

$$p(x) = \sum_{S \subset [n]} \widehat{p}(S) \prod_{i \in S} x(i),$$

the influence of coordinate j is

$$\text{Inf}_j[p] = \sum_{S \ni j} \widehat{p}(S)^2, \quad \text{MaxInf}[p] = \max_j \text{Inf}_j[p].$$

For $\mathbf{i} = (i_1, \dots, i_d) \in [n+1]^d$, the set $S_{\mathbf{i}} \subset [n]$ contains the indices in $[n]$ that occur an odd number of times in $\mathbf{i}$; the extra coordinate $n+1$ is ignored.

The source's Boolean-behavior tuples consist of unit vectors u, v and contractions $A(k)$ such that the scalar $\langle u, A(i_1) \cdots A(i_d)v \rangle$ depends only on $S_{\mathbf{i}}$. The only property used below is the immediate bound

$$|\langle u, A(i_1) \cdots A(i_d)v \rangle| \leq 1.$$

Lemma 1. *Let u, v be unit vectors and $A(1), \dots, A(n+1)$ contractions on a real Hilbert space. Then, for every $\mathbf{i} \in [n+1]^d$,*

$$|\langle u, A(i_1) \cdots A(i_d)v \rangle| \leq 1.$$

Proof. The product $A(i_1) \cdots A(i_d)$ is a contraction. Hence

$$|\langle u, A(i_1) \cdots A(i_d)v \rangle| \leq \|u\| \|A(i_1) \cdots A(i_d)v\| \leq 1.$$

□

Counterexample

Theorem 1. *Under the usual reading that the displayed $\text{poly}(d, \text{MaxInf}[p])$ is a fixed finite nonzero denominator depending only on d and $\text{MaxInf}[p]$, Question 4.5 is false without a normalization hypothesis on p .*

Proof. Fix $d \geq 1$ and $n \geq 1$. Let $D(d, t)$ denote any proposed finite nonzero denominator depending only on d and $t = \text{MaxInf}[p]$. Choose $\varepsilon > 0$ such that $D(d, \varepsilon^2)$ is finite and nonzero, and then choose

$$M > |D(d, \varepsilon^2)|.$$

Consider the degree-one polynomial

$$p(x) = M + \varepsilon x(1), \quad x \in \{-1, 1\}^n.$$

Its only nonzero Fourier coefficients are

$$\widehat{p}(\emptyset) = M, \quad \widehat{p}(\{1\}) = \varepsilon,$$

and therefore

$$\text{MaxInf}[p] = \text{Inf}_1[p] = \varepsilon^2.$$

Choose any $\mathbf{i} \in [n+1]^d$ with $S_{\mathbf{i}} = \emptyset$, for instance $\mathbf{i} = (n+1, \dots, n+1)$. If the conclusion of Question 4.5 held with denominator $D(d, \text{MaxInf}[p])$, then some Boolean-behavior tuple (u, v, A) would satisfy

$$\langle u, A(i_1) \cdots A(i_d)v \rangle = \frac{\hat{p}(\emptyset)}{D(d, \varepsilon^2)} = \frac{M}{D(d, \varepsilon^2)}.$$

The right-hand side has absolute value greater than 1, while the lemma shows that every such correlation has absolute value at most 1. This is a contradiction. $\square$

Verification Notes

The proof uses no computational input. It only uses the contraction bound implicit in the definition of $\mathcal{B}\mathcal{B}^d$ and the elementary fact that influences ignore the constant Fourier coefficient.

The counterexample is stable under replacing the constant polynomial by a nonconstant linear polynomial $M + \varepsilon x(1)$; hence it is not merely a division-by-zero issue at $\text{MaxInf}[p] = 0$.

Novelty Check

The cheap local indexes were searched for arXiv:2304.06713, the paper title, “Fourier completely bounded”, “MaxInf”, and close operator-space keywords. No exact prior packet was found. A bounded web search on 2026-07-03 for “Fourier completely bounded Question 4.5”, “Influences of Fourier Completely Bounded Polynomials Question 4.5”, and close variants found the source arXiv record but no later discussion of this normalization issue.

Limitations and Human-Review Recommendation

This packet answers only the literal unnormalised wording of Question 4.5. It does not refute Conjecture 1.5 in the source, the Aaronson–Ambainis conjecture, or any version of Question 4.5 that assumes $\|p\|_{\text{fcb}, d} \leq 1$ or allows the denominator to depend on $|\hat{p}(\emptyset)|$.

Human review should check whether the intended convention around $\text{poly}(d, \text{MaxInf}[p])$ silently included such a normalization. If so, this packet should be classified as a correction to the printed auxiliary formulation rather than as progress on the intended normalized problem.

References

- [1] Francisco Escudero Gutierrez, *Influences of Fourier Completely Bounded Polynomials and Classical Simulation of Quantum Algorithms*, arXiv:2304.06713, 2023.